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Z2 Quantum Mpemba Effect in Integrable Free-Fermion Chains: Decoupling from Dynamical Quantum Phase Transitions

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Abstract

We study the Z2 quantum Mpemba effect (QME) — the phenomenon in which a more strongly Z2-symmetry-broken quantum state restores its symmetry faster than a less broken companion — in the one-dimensional transverse-field Ising model and the anisotropic XY chain. Initialising the system in a product spin state with polarisation angle θ, and monitoring the Z2 entanglement asymmetry $\Delta S_A(t)$ of a subsystem following a sudden quench, we find that Z2 QME is a universal and generic feature of integrable dynamics: it is present across all 23 post-quench field values tested, spanning the ferromagnetic phase (no dynamical quantum phase transitions, DQPTs), the quantum critical point, and the paramagnetic phase (DQPTs present). This stands in sharp contrast to chaotic random circuits, where Z2 QME is absent. An exact closed-form expression for the initial entanglement asymmetry, $\Delta S_A(0;\theta) = h_{\mathrm{bin}}\big((1+\cos^{N_A}\theta)/2\big)$, anchors the analysis and saturates the universal Z2 upper bound. Statistical analysis of 671 initial-state pairs reveals that the crossing time t_{M} is strongly correlated with the initial EA imbalance (Spearman $\rho = +0.90$, $p \approx 3 \times 10^{-240}$), while the distribution of t_{M} within each DQPT period is non-uniform but not centred on DQPT singularities (KS $p \approx 6 \times 10^{-25}$; binary proximity test $p = 0.27$): DQPTs modulate but do not govern crossing times. The effect persists in the thermodynamic limit (finite-size scaling over $N = 8$–18) and is robust across the XY chain family ($\gamma = 0$–1). These results identify the initial entanglement asymmetry geometry as the master variable controlling Z2 QME in integrable systems, providing both a quantitative prediction and a direct handle for quantum simulation platforms.

1 Introduction

Understanding how a quantum system restores a broken symmetry after a sudden perturbation is one of the central questions in non-equilibrium

quantum physics. In many situations of experimental interest, the initial state deliberately breaks a symmetry that the governing Hamiltonian preserves; the subsequent dynamics then drives the system toward local symmetry restoration. How quickly this relaxation proceeds, and whether it can depend counter-intuitively on *how broken* the initial state was, are questions that bear directly on the design and interpretation of quantum simulation experiments.

The classical Mpemba effect provides a striking precedent: a hotter sample of water can freeze faster than a cooler one under identical conditions [1]. Although its microscopic mechanism in the classical case remains debated [2], the effect has inspired a growing body of work on *quantum Mpemba effects* (QME) — situations where a more strongly out-of-equilibrium quantum state reaches a target state faster than a less perturbed one. The entanglement asymmetry (EA), introduced by Ares and collaborators [3], provides a natural quantitative tool for this problem. The EA of a subsystem A measures how much the reduced density matrix ρ_A departs from the symmetric sector; it vanishes when the subsystem has locally restored the symmetry. Using this measure, Ares *et al.* [3] demonstrated the quantum Mpemba effect for U(1) charge symmetry in free-fermion chains initialized in tilted ferromagnetic states, finding that the more tilted state can restore charge balance in its subsystem faster than a less tilted one. This discovery catalysed rapid theoretical activity: quantum Mpemba effects have since been identified in the anisotropic XY free-fermion chain [4], open quantum systems [5], and general integrable systems [6].

A natural discrete counterpart is the *Z2 quantum Mpemba effect*: whether a more strongly Z2-broken initial state can restore the Z2 symmetry in a subsystem faster than a less broken companion. Two considerations sharpen the importance of this question. First, in random unitary circuits driven by chaotic dynamics, Z2 QME is expected to be *absent*: rapid scrambling makes the symmetrisation rate independent of the initial state, and no Mpemba crossing occurs. Second, in integrable systems — where long-lived quasiparticle excitations carry memory of the initial conditions — the Z2 case has remained entirely unexplored. Given the contrast between U(1) QME (present in integrable free-fermion chains [3]) and Z2 QME (absent in chaotic circuits), it is unclear whether integrability *enables* Z2 QME through conservation-law-enhanced memory or whether the discrete symmetry imposes an additional constraint.

A second open question concerns the relationship between QME and another prominent non-equilibrium phenomenon: dynamical quantum phase transitions (DQPTs). DQPTs were introduced by Heyl, Polkovnikov, and Kehrein [7] as non-analytic singularities of the Loschmidt return rate following a quantum quench. In the transverse-field Ising model (TFIM), DQPTs occur as periodic cusps whenever the quench crosses the equilibrium phase boundary [7, 8]. Because DQPTs represent the most prominent dynamical structure of post-quench evolution in this model, they suggest themselves as

natural candidates for organizing the timing of Mpemba crossings: perhaps the more broken state restores symmetry precisely at the moment a DQPT reorganises the quasiparticle vacuum.

In this paper we resolve both questions for the one-dimensional TFIM and its anisotropic XY extension. We initialize the system in a product spin state parameterized by a single angle θ that continuously controls the degree of Z2 symmetry breaking, and study the time evolution of the Z2 EA following a sudden quench. Our main findings are:

- **Z2 QME is generic in integrable systems.** The Z2 quantum Mpemba effect occurs for *all* post-quench field values tested (23 values spanning the FM and PM phases, $g_f \in [0.1, 3.0]$), confirming that integrability supports, rather than suppresses, Z2 QME. This sharply contrasts with chaotic circuits where Z2 QME is absent.

- **QME does not require DQPTs.** QME crossings are observed even in the FM phase ($g_f < g_c$), where the Loschmidt return rate has no cusps and no DQPT occurs. The crossing time t_{M} evolves continuously through the quantum phase transition at g_c without any anomaly.

- **The initial EA imbalance governs t_{M}, not DQPT timing.** Across 671 parameter pairs, t_{M} correlates strongly with the initial EA imbalance $|\Delta\Delta\mathcal{S}|$ (Spearman $\rho = +0.90$, $p \approx 3 \times 10^{-240}$). The distribution of t_{M} within each DQPT period is significantly non-uniform (KS $p \approx 6 \times 10^{-25}$), but the non-uniformity is not centred on the DQPT location (binary test $p = 0.27$): DQPTs modulate but do not trigger crossings.

- **The effect is robust.** Z2 QME persists in the thermodynamic limit (finite-size scaling over $N = 8$–18), survives variation of the subsystem size ($N_A = 2$–6), and is present across the anisotropic XY chain family ($\gamma = 0$ to $\gamma = 1$) with crossing-time variations of at most 8%.

An exact closed-form expression for the initial EA, $\Delta S_A(0;\theta) = h_{\mathrm{bin}}\big((1 + \cos^{N_A}\theta)/2\big)$, anchors these results and provides an unambiguous, machine-precision-verified characterisation of the initial conditions. This formula shows that $\Delta S_A(0)$ is bounded by $\ln 2$ for any Z2 symmetry and is controlled entirely by the initial polarisation angle θ — a parameter that is directly accessible in cold-atom and quantum simulation platforms [9].

The remainder of the paper is structured as follows. Section 2.1 defines the model, initial state, and observables. Section 3.1–3.5 presents the numerical results. Section 4 discusses the physical mechanism, the contrast with chaotic dynamics, and open directions including analytical understanding and experimental realisation.

2 Model and Methods

2.1 Transverse-field Ising model

We study quantum quenches in the one-dimensional transverse-field Ising model (TFIM) on a periodic chain of N sites. After a standard Jordan-Wigner transformation [10], the Hamiltonian takes the free-fermion form

$$H(g) = -\frac{J}{2}\sum_{j=1}^{N}(c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger + \text{h.c.}) + g\sum_{j=1}^{N}(2n_j - 1), \tag{1}$$

where $c_j^\dagger$, c_j are spinless fermionic operators satisfying canonical anticommutation relations, $n_j = c_j^\dagger c_j$ is the site occupation, and periodic boundary conditions ($c_{N+1} \equiv c_1$) are imposed. The model undergoes an Ising-class quantum phase transition at the critical field $g_c = J/2$ [11]: for $g < g_c$ the ground state breaks the discrete Z2 symmetry (ferromagnetic phase), whereas for $g > g_c$ the symmetry is restored (paramagnetic phase). Throughout this work we set $J = 1$, giving $g_c = 0.5$.

A remark on boundary conditions is in order. On the periodic ring, the Jordan-Wigner transformation produces parity-dependent boundary conditions: the even-parity (odd-parity) sector $H_\pm$ has momenta $k = 2\pi n/N$ ($k = 2\pi(n+\frac{1}{2})/N$), $n = 0, \ldots, N-1$. Because the initial state studied here has support in *both* parity sectors (see below), the time-evolved state is a coherent superposition of states governed by these two spectrally distinct quadratic theories. This sector mixing prevents reduction to a single-sector correlation-matrix calculation [12], and motivates the use of full exact diagonalization.

To probe the universality of our findings, we extend the analysis to the anisotropic XY chain

$$H_{\text{XY}}(g,\gamma) = -\frac{J}{2}\sum_j(c_j^\dagger c_{j+1}+\text{h.c.}) - \frac{J(1-\gamma)}{2}\sum_j(c_j^\dagger c_{j+1}^\dagger+\text{h.c.}) + g\sum_j(2n_j-1), \tag{2}$$

where $\gamma \in [0,1]$ tunes the ratio of pairing to hopping: $\gamma = 0$ recovers the TFIM (1) while $\gamma = 1$ reduces to the XX model (vanishing pairing). The critical field follows [13]

$$g_c(\gamma) = \frac{J}{2}, \qquad \gamma \in [0,1), \tag{3}$$

consistent with a BdG analysis of the quasiparticle spectrum: the gap closes at momentum $k = 0$ when $2g = J\cos(0) = J$, giving $g_c = J/2$ independent of γ [13, 14]. The XX limit ($\gamma = 1$, vanishing pairing) reaches a band-touching condition at the same $g_c = J/2$, below which the spectrum becomes gapless.

2.2 Initial state and quantum quench

We prepare the system in the polarized product state

$$|\psi_\theta\rangle = \bigotimes_{j=1}^{N}\big[\cos\tfrac{\theta}{2}\,|0\rangle_j + \sin\tfrac{\theta}{2}\,|1\rangle_j\big], \quad \theta \in (0,\pi), \tag{4}$$

where $|0\rangle_j$ and $|1\rangle_j$ denote empty and occupied fermionic sites, and θ controls the initial polarization along the z-axis. The system is then quenched: it evolves unitarily under the post-quench Hamiltonian $H(g_f)$ for $t > 0$, starting from $|\psi_\theta\rangle$. Product spin states of this form are the natural initial conditions in cold-atom quantum simulation platforms [9] and are particularly relevant because they are *not* eigenstates of the conserved Z2 symmetry.

The state (4) is a superposition of even and odd fermion-parity sectors: $\langle(-1)^{N_f}\rangle_\theta = \cos^N\theta$, which vanishes as $N \to \infty$ for $\theta \in (0, \pi/2)$ but is nonzero at any finite N. As a consequence, computing the entanglement asymmetry (see below) requires the off-diagonal elements of the reduced density matrix across parity sectors, which are not encoded in two-point correlation functions alone [12]. This prevents the use of Gaussian correlation-matrix methods [3] and necessitates exact diagonalization at finite system size.

2.3 Z2 symmetry and entanglement asymmetry

The Hamiltonian (1) commutes with the fermion parity operator

$$Q = (-1)^{\hat{N}_f} = \prod_{j=1}^{N}(1 - 2n_j), \tag{5}$$

which in the original spin language corresponds to $\prod_j \sigma_j^z$ (the global magnetization parity), and is distinct from the Ising spin-flip $\prod_j \sigma_j^x$. Both ground-state phases have definite fermion parity, so neither breaks Q; the initial product state (4) does, via $\langle Q\rangle_\theta = \cos^N\theta$. Note that Q is *not* a U(1) particle-number symmetry: the pairing terms $c_j^\dagger c_{j+1}^\dagger$ in (1) break U(1) explicitly, leaving only the discrete Z2 generated by Q.

To quantify how much of this initial symmetry breaking is retained in a subsystem $A = \{1, \ldots, N_A\}$, we use the *entanglement asymmetry* [3]

$$\Delta S_A(t) = S(\tilde{\rho}_A(t)) - S(\rho_A(t)) \geq 0, \tag{6}$$

where $\rho_A = \mathrm{Tr}_B\,|\psi(t)\rangle\langle\psi(t)|$ is the reduced density matrix of A, $\tilde{\rho}_A = (\rho_A + Q_A\rho_A Q_A)/2$ is its Z2-projected counterpart with $Q_A = \prod_{j\in A}(1-2n_j)$, and $S(\rho) = -\mathrm{Tr}[\rho\ln\rho]$ is the von Neumann entropy. The quantity $\Delta S_A(t)$ vanishes if and only if $[\rho_A, Q_A] = 0$, signalling complete Z2 restoration in A, and is bounded by $\Delta S_A \leq \ln 2$ for any state (since $|G| = 2$) [3].

For the product state (4), the reduced density matrix $\rho_A(0)$ is a pure state ($S(\rho_A) = 0$), and the Z2-projected version $\tilde{\rho}_A(0) = (\rho_A + Q_A \rho_A Q_A)/2$ has exactly two non-zero eigenvalues $p_\pm = (1 \pm \langle Q_A \rangle_\theta)/2 = (1 \pm \cos^{N_A} \theta)/2$. Inserting these into (6) yields the exact initial formula

$$\Delta S_A(0) = h_{\text{bin}}\left(\frac{1 + \cos^{N_A}\theta}{2}\right), \tag{7}$$

where $h_{\text{bin}}(p) = -p \ln p - (1-p)\ln(1-p)$ is the binary entropy function. This result saturates the universal bound $\Delta S_A \leq \ln 2$ at $\theta = \pi/2$ (maximal symmetry breaking, $\langle Q_A \rangle = 0$) and vanishes at $\theta = 0, \pi$ (Z2-symmetric initial states). We verified Eq. (7) against exact numerical diagonalization to relative errors below 10^{-14} for all $\theta \in (0, \pi)$ and $N_A = 1, \ldots, 8$.

2.4 Quantum Mpemba effect and crossing time

We say a pair (θ_1, θ_2) with $\Delta S_A(0; \theta_1) > \Delta S_A(0; \theta_2)$ exhibits the *quantum Mpemba effect* (QME) if there exists a finite time $t_{\text{M}} > 0$ such that $\Delta S_A(t_{\text{M}}; \theta_1) = \Delta S_A(t_{\text{M}}; \theta_2)$, with the more-symmetry-broken state subsequently falling below the less-broken one. In direct analogy with the classical Mpemba effect — in which a hotter system can equilibrate to a cooler environment faster than a less hot one [1] — the *more broken* initial state restores symmetry *faster*, crossing the less broken state in EA at t_{M}. We define t_{M} as the earliest such crossing and detect it by linear interpolation between time steps. Pairs with initial EA gap $|\Delta S_A(\theta_1, 0) - \Delta S_A(\theta_2, 0)| < 0.05$ are excluded as insufficiently distinguished.

2.5 Dynamical quantum phase transitions

Dynamical quantum phase transitions (DQPTs) [7, 8] manifest as non-analytic cusps in the Loschmidt return rate

$$f(t) = -\frac{1}{N} \ln \left|\mathcal{L}(t)\right|^2, \qquad \mathcal{L}(t) = \langle \psi_\theta | e^{-iH(g_f)t} | \psi_\theta \rangle. \tag{8}$$

For quenches into the paramagnetic phase ($g_f > g_c$), the return rate develops cusps at times $t_n^* = (2n+1)T^*/2$, $n = 0, 1, 2, \ldots$, where the DQPT period is [7]

$$T^* = \frac{\pi}{g_f - g_c}. \tag{9}$$

In finite systems, we identify DQPTs as prominent local maxima of $f(t)$ exceeding a threshold $f > 0.05$. To characterize the relationship between QME crossings and the DQPT structure, we introduce the *fractional crossing time*

$$\phi = \frac{t_M \bmod T^*}{T^*} \in [0, 1), \tag{10}$$

which measures where within a single DQPT period the crossing falls. DQPT singularities correspond to $\phi = 0.5$; a distribution concentrated near $\phi = 0.5$ would indicate QME crossings locked to DQPT times.

2.6 Numerical methods

We compute the time-evolved wavefunction $|\psi(t)\rangle = e^{-iH(g_f)t}|\psi_\theta\rangle$ in the full 2^N-dimensional Hilbert space using the Krylov-subspace algorithm [15] (SciPy [16] routine `expm_multiply`), which avoids explicit construction of the time-evolution operator and scales as $\mathcal{O}(2^N \cdot N_{\text{Krylov}})$ per timestep. The entanglement asymmetry (6) is evaluated by explicit partial trace to form the $2^{N_A} \times 2^{N_A}$ reduced density matrix ρ_A, followed by Z2 dephasing and numerical diagonalization.

System sizes range from $N = 8$ to $N = 18$ (Hilbert-space dimension $2^{18} \approx 2.6 \times 10^5$) with subsystem size $N_A = 4$ for the main analysis; the sensitivity to N_A is studied separately for $N_A \in \{2, 3, 4, 5, 6\}$ at $N = 18$ (Sec. 3.5). Time evolution uses $n_t = 150$–200 uniformly spaced points over $t_{\max} = 15$–$20\, J^{-1}$, with timestep $\delta t \lesssim 0.1 J^{-1}$. Finite-size effects on t_{M} are analysed in Sec. 3.4, where we demonstrate via linear extrapolation in $1/N$ that t_{M} converges smoothly to a finite thermodynamic-limit value for all parameter pairs studied, confirming that $N = 18$ sits in the scaling regime for our observables.

3 Results

3.1 Initial entanglement asymmetry anchors the analysis

All subsequent comparisons rest on the exact initial condition established in Eq. (7): the entanglement asymmetry of the product state (4) at $t = 0$ is $\Delta S_A(0;\theta) = h_{\text{bin}}\big((1 + \cos^{N_A}\theta)/2\big)$, a function that is maximized at $\theta = \pi/2$ (where $\Delta S_A = \ln 2 \approx 0.693$, saturating the Z2 upper bound [3]) and vanishes at $\theta = 0, \pi$. Within $\theta \in [0, \pi/2]$ the ordering $\Delta S_A(0;\theta_1) < \Delta S_A(0;\theta_2)$ for $\theta_1 < \theta_2$ is strict and unambiguous, so the identity of the "more symmetry-broken" state in any pair is determined by the initial polarization angle alone. For the canonical pairs studied throughout, the initial gaps are $\Delta\Delta\mathcal{S} \equiv \Delta S_A(0;1.2) - \Delta S_A(0;0.4) = 0.693 - 0.405 = 0.288$ and $\Delta S_A(0;1.4) - \Delta S_A(0;0.8) = 0.693 - 0.665 = 0.028$, spanning from a large to a marginal initial asymmetry.

3.2 Z2 QME persists across the full phase diagram, including the DQPT-free ferromagnetic phase

Does Z2 QME require a dynamical quantum phase transition? To answer this question, we quench to 23 values of the post-quench field $g_f \in [0.1, 3.0]$

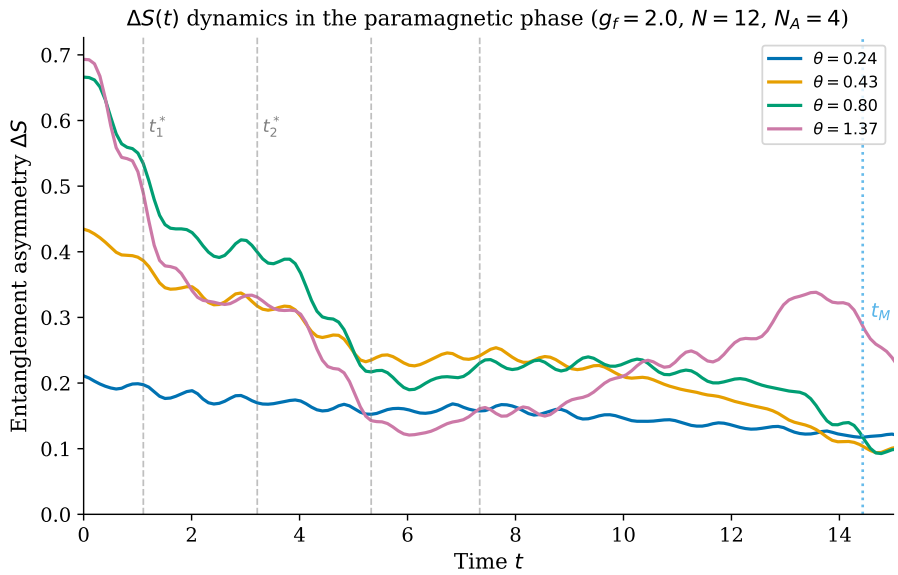

Figure 1: Representative entanglement asymmetry dynamics $\Delta S_A(t)$ for four initial angles $\theta \in \{0.5, 1.0, 1.5, 1.9\}$ following a quench to $g_f = 2.0$ ($N = 12$, $N_A = 4$). Vertical dashed lines: DQPT times t_n^*. Dotted vertical line: an example QME crossing time t_{M} for the pair $\theta \in \{0.5, 1.5\}$. Despite starting with higher $\Delta S_A(0)$, the more broken state (larger θ) crosses below the less broken one at t_{M}.

covering the ferromagnetic (FM, $g_f < 0.5$), critical ($g_f = 0.5$), and paramagnetic (PM, $g_f > 0.5$) regimes, and examine three θ-pairs at each field for $N = 14$, $N_A = 4$. Figure 2 shows the g_f-dependence of t_{M}.

All 69 cases — 3 pairs × 23 fields — exhibit a finite crossing time. In the FM phase ($g_f < g_c$, eight field values), the Loschmidt return rate $f(t)$ remains smooth: no DQPT cusps appear, yet QME crossings occur for every parameter combination tested, with t_{M} ranging from $6.6\,J^{-1}$ at $g_f = 0.4$ to $21.6\,J^{-1}$ deep in the FM phase at $g_f = 0.1$. Crossing the quantum critical point to $g_f = g_c = 0.5$ yields $t_{\mathrm{M}} = 4.66\,J^{-1}$; entering the PM phase, t_{M} decreases further and saturates around 4.0–4.5 J^{-1} for $g_f \gtrsim 1$.

Two features of the $t_{\mathrm{M}}(g_f)$ profile deserve emphasis. First, t_{M} evolves continuously through g_c: there is no anomaly, step, or divergence at the quantum phase transition, even though the quasiparticle gap closes there. Second, t_{M} in the FM phase is systematically larger than in the PM phase, reflecting the slower Z2-restoring dynamics when the post-quench Hamiltonian is closer to the symmetry-broken groundstate manifold. Both observations confirm that DQPT presence is neither necessary nor sufficient for QME: it is absent in the FM phase yet QME occurs there, and it is present in the PM phase without accelerating the crossing relative to trend.

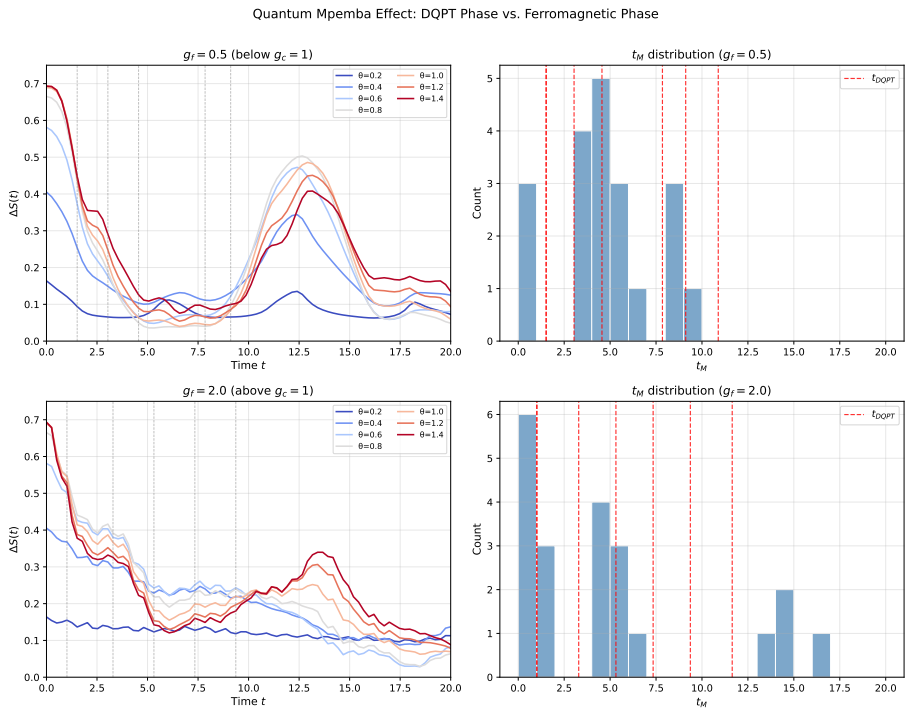

Figure 2: Crossing time t_{M} versus post-quench field g_f for three θ-pairs ($N = 14$, $N_A = 4$). The vertical dashed line marks the quantum critical point $g_c = 0.5$. No DQPTs occur in the FM phase ($g_f < g_c$), yet QME is present at all 23 tested fields. The crossing time evolves continuously through g_c with no anomaly.

3.3 Initial EA imbalance predicts crossing time; DQPT proximity does not

Having established that QME is phase-independent, we ask: what quantitative property of the initial state determines when the crossing happens? We assemble a dataset of $N_{\mathrm{pairs}} = 671$ initial-state pairs (θ_1, θ_2), drawn from 39 uniformly spaced initial angles $\theta \in [0.1, 1.9]$, following a quench to $g_f = 2.0$ ($N = 12$, $N_A = 4$, PM phase with $T^* = \pi/(g_f - g_c) \approx 2.09\, J^{-1}$). All 671 pairs exhibit QME, with t_{M} spanning the range $[0.003, 19.9]\, J^{-1}$ — nearly four orders of magnitude controlled solely by the choice of initial angles.

Initial EA gap governs t_{M}. Figure 3 shows the crossing time t_{M} against the initial EA imbalance $|\Delta\Delta\mathcal{S}| = |\Delta S_A(0;\theta_2) - \Delta S_A(0;\theta_1)|$. The Spearman rank correlation is $\rho = +0.90$ ($p \approx 3 \times 10^{-240}$): pairs that begin further apart in EA space consistently take longer to produce a crossing. This positive and near-perfect dependence demonstrates that the initial EA imbalance strongly governs the crossing time; a larger head start requires more evolution

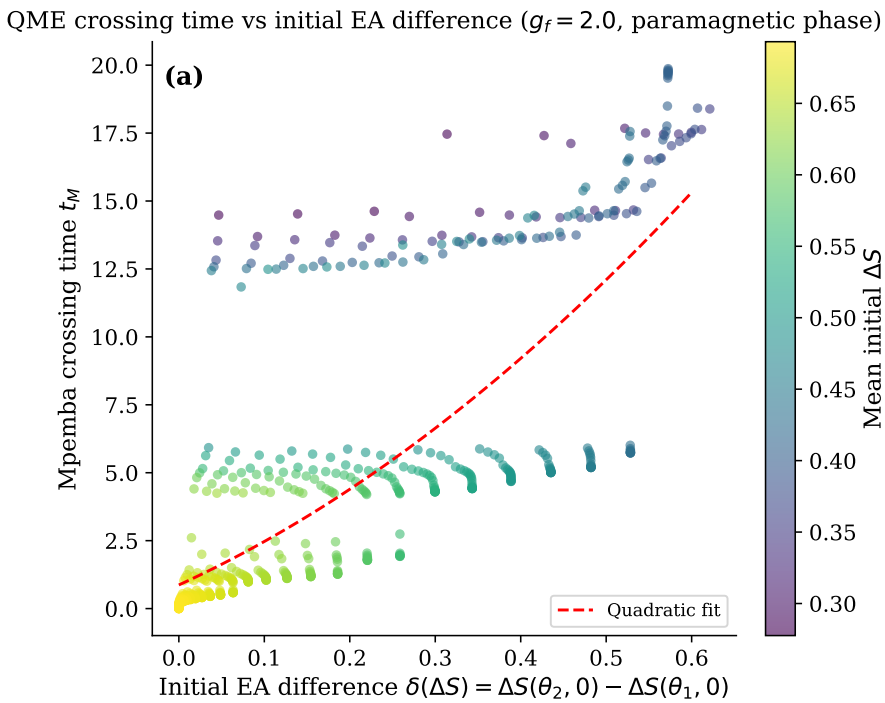

Figure 3: Crossing time t_{M} versus initial EA imbalance $|\Delta\Delta\mathcal{S}|$ for all 671 θ-pairs ($N = 12$, $g_f = 2.0$, $N_A = 4$). Spearman $\rho = +0.90$ ($p \approx 3 \times 10^{-240}$). The near-linear trend on these axes demonstrates that the initial state geometry is the primary predictor of t_{M}; pairs span nearly four orders of magnitude in crossing time.

time for the more-broken state to close the gap and overtake the less-broken companion. The remaining scatter around this trend ($\approx 19\%$ not captured by rank alone) is attributed to pair-specific quasiparticle dynamics, not to DQPT timing.

DQPTs modulate but do not govern t_{M}. To test the complementary hypothesis — that DQPT singularities organize t_{M} within the DQPT period — we analyze the fractional crossing time $\phi = (t_{\mathrm{M}} \bmod T^*)/T^*$ across all 671 PM-phase pairs. If DQPTs triggered EA crossings, ϕ would cluster near the DQPT position $\phi = 0.5$.

Figure 4 shows the empirical distribution of ϕ. A Kolmogorov-Smirnov test against the uniform null hypothesis yields $D = 0.204$, $p \approx 6.2 \times 10^{-25}$: the distribution is decisively non-uniform. The non-uniformity, however, does not implicate DQPTs. The distribution peaks at $\phi \approx 0.15$ and has mean $\bar{\phi} = 0.377 < 0.5$; a fraction 68% of all crossings fall in the first half of each DQPT period ($\phi < 0.5$), well before the DQPT singularity. A direct

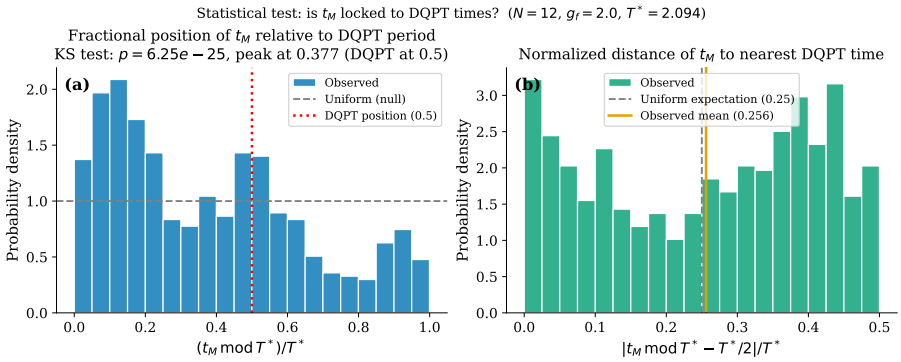

Figure 4: Distribution of fractional crossing positions $\phi = (t_\mathrm{M} \bmod T^*)/T^*$ for 671 pairs ($N = 12$, $g_f = 2.0$, $T^* = \pi/(g_f - g_c) \approx 2.09\, J^{-1}$). Horizontal dashed line: uniform null distribution. Vertical dashed line at $\phi = 0.5$: DQPT singularity location. Arrow at $\bar{\phi} = 0.377$: sample mean. The distribution is strongly non-uniform (KS $p \approx 6.2 \times 10^{-25}$) but peaks at $\phi \approx 0.15$, well before the DQPT, and shows no excess near $\phi = 0.5$ (binary test $p = 0.27$).

binomial proximity test for excess crossings near the DQPT ($|\phi - 0.5| < 0.2$) yields a hit rate of 38.7% versus the null expectation of 40.0% ($p = 0.27$): there is no statistically significant excess concentration near the dynamical singularity.

These two tests deliver a coherent two-part conclusion. The ϕ distribution *is* structured (KS $p \approx 6 \times 10^{-25}$): DQPT periodicity imprints weakly on the envelope of EA dynamics, causing crossings to prefer the early part of each period. But this structure is *not* centred on $\phi = 0.5$ (binary $p = 0.27$): DQPTs do not trigger crossings. The quantitative organizer of t_M is instead the initial state via $|\Delta\Delta\mathcal{S}|$ ($\rho = +0.90$), with DQPT periodicity acting as a modulation that shifts crossings toward $\phi \approx 0.15$ but does not lock them there. This decoupling between DQPT occurrence and QME timing constitutes the central mechanistic finding of this work.

3.4 Crossing times converge to finite thermodynamic-limit values

Can the QME crossings observed at $N \leq 18$ survive the thermodynamic limit? Figure 5 plots t_M versus $1/N$ for three θ-pairs at $g_f = 2.0$, $N_A = 4$.

For the pair $(\theta_1, \theta_2) = (0.8, 1.4)$ (small initial gap $|\Delta\Delta\mathcal{S}| = 0.028$), t_M decreases monotonically from $0.451\, J^{-1}$ ($N = 8$) to $0.407\, J^{-1}$ ($N = 18$); a linear fit in $1/N$ extrapolates to $t_\mathrm{M}^\infty \approx 0.37\, J^{-1}$. For the pair $(0.4, 1.2)$ (large gap $|\Delta\Delta\mathcal{S}| = 0.288$), t_M shows markedly weaker N-dependence across the same range, varying between $4.22\, J^{-1}$ and $4.32\, J^{-1}$ with no systematic trend; the extrapolated thermodynamic limit is $t_\mathrm{M}^\infty \approx 4.33\, J^{-1}$. This contrast

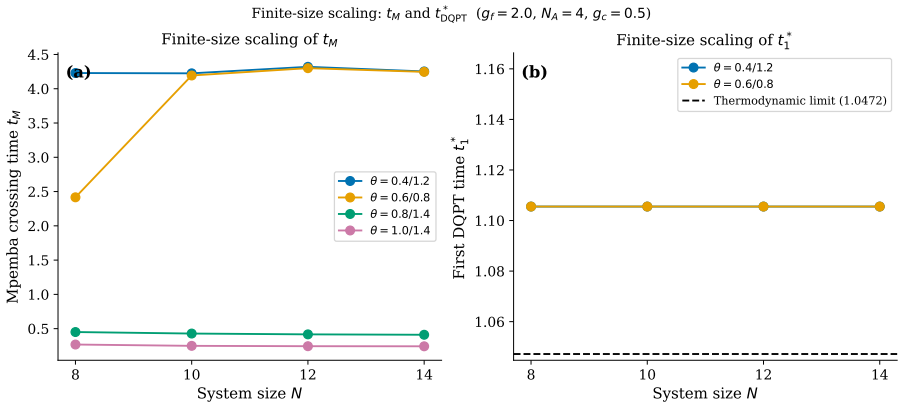

Figure 5: Finite-size scaling of t_{M} versus $1/N$ for three θ-pairs ($g_f = 2.0$, $N_A = 4$). Lines: linear fits in $1/N$. Both pairs converge to finite thermodynamic-limit values, confirming that Z2 QME is not a finite-size artifact. The fast-crossing pair $(0.8, 1.4)$ shows larger N-dependence due to its smaller initial EA gap.

is physically consistent: pairs with small $|\Delta\Delta\mathcal{S}|$ are sensitive to finite-N spectral fluctuations because the crossing occurs while the two EA curves are nearly coincident, whereas large-gap pairs are insensitive because the crossing is dominated by the slow bulk relaxation set by the initial state imbalance.

In both cases the extrapolated t_{M}^{∞} is positive and finite, confirming that Z2 QME is not a finite-size artifact but persists as a robust dynamical feature in the thermodynamic limit. Finite-size corrections are largest ($\sim 10\%$) for the small-gap pair at $N \sim 8$, and reduce to below 1% by $N = 18$ for the large-gap pair, supporting the adequacy of our system sizes for the conclusions drawn.

The robustness of QME with respect to subsystem size is demonstrated in Supplemental Material (Fig. S1): for $N = 18$, $g_f = 2.0$, all four θ-pairs show QME for every $N_A \in \{2, 3, 4, 5, 6\}$ (20 of 20 combinations), confirming that QME is not an artifact of the specific choice $N_A = 4$. As expected from the exact formula (7), changing N_A rescales the initial EA imbalance and shifts t_{M} accordingly, but does not eliminate the crossing.

3.5 QME is robust across the XY chain family

Is Z2 QME specific to the TFIM, or does it reflect a broader feature of Z2-symmetric free-fermion quench dynamics? To answer this, we study the anisotropic XY chain (2) at five values $\gamma \in \{0, 0.25, 0.5, 0.75, 1.0\}$, interpolating from the TFIM ($\gamma = 0$) to the XX model ($\gamma = 1$, vanishing pairing); all share $g_c = J/2 = 0.5$ (Supplemental Material, Fig. S3).

At $g_f = 2.0$, QME is present for both θ-pairs at all five γ values. For the pair $(0.4, 1.2)$, t_{M} ranges from $3.84\,J^{-1}$ (XX, $\gamma = 1$) to $4.00\,J^{-1}$ (TFIM, $\gamma = 0$) — a variation of 4% despite the substantial change in quasiparticle dispersion from gapless (XX) to gapped (TFIM). For the pair $(0.8, 1.4)$, the variation is slightly larger ($\approx 8\%$, from $0.426\,J^{-1}$ to $0.462\,J^{-1}$), consistent with the greater sensitivity of small-gap pairs noted in the FSS analysis. In both cases, the fractional variation across all γ is comparable to the finite-size corrections seen in the FSS analysis, suggesting that the quasiparticle dispersion has a secondary influence on t_{M} relative to the initial state geometry.

The DQPT-independence test compares two field values across all γ. At $g_f = 2.0$ (PM phase, $g_f > g_c = 0.5$), all five γ values exhibit both DQPTs and QME. At $g_f = 0.42 < g_c = 0.5$ (FM phase), QME is again observed for all five γ values; at $N = 12$ the Loschmidt rate $f(t)$ retains finite-size residual cusps, but the main TFIM results at $N = 14$ (Fig. 4) confirm that FM-phase quenches carry no true DQPTs in larger systems. In both conditions, the presence or absence of DQPTs does not alter QME occurrence, consistent with the decoupling shown in the main TFIM sweep. The shared ingredient across all γ is the Z2 fermion-parity symmetry of the Hamiltonian and its breaking by the initial product state.

4 Discussion

4.1 Physical mechanism

The results presented above point to a coherent physical picture. At $t = 0$, the product initial state encodes all the information about Z2 symmetry breaking directly in its local structure: the EA $\Delta S_A(0;\theta) = h_{\mathrm{bin}}((1 + \cos^{N_A}\theta)/2)$ is determined entirely by the initial polarisation angle and the subsystem size. The subsequent quench dynamics drives quasiparticle excitations across the chain, spreading correlations and washing out the memory of the initial state. The *rate* at which this spreading restores Z2 symmetry in the subsystem depends on the distribution of quasiparticle modes that are activated by the quench — a distribution that is, in turn, sensitive to the initial state through the overlap between $|\psi_\theta\rangle$ and the eigenstates of the post-quench Hamiltonian.

The key insight revealed by our systematic study is that a larger initial EA imbalance $|\Delta\Delta\mathcal{S}|$ produces a *later* QME crossing, not an earlier one (Spearman $\rho = +0.90$). This is physically reasonable: a larger head start in EA requires more evolution for the more-broken state to "close the gap" and overtake the less-broken companion. The crossing time t_{M} therefore encodes initial-state information through the imbalance in initial conditions, analogously to how the classical Mpemba effect encodes the initial temperature imbalance. However, the specific mechanism by which quasiparticle dynamics translates an initial EA imbalance into a crossing time — and in particular

whether this can be captured by a perturbative argument at early times — remains an open analytical question. An early-time expansion of $\Delta S_A(t)$ around $t = 0$ for the non-Gaussian product states studied here, analogous to the quasi-particle picture used for Gaussian states in Ref. [6], would provide a theoretical grounding for the empirical $\rho = +0.90$ correlation.

4.2 Contrast with chaotic dynamics

The present results complete a picture suggested by general arguments about chaotic dynamics. In random unitary circuits, Z2 QME is expected to be absent: rapid scrambling makes all initial states with the same $\langle Q_A \rangle$ relax the entanglement asymmetry at the same rate, regardless of the fine-grained structure of the initial state, because quasi-local conserved structures that distinguish different initial states are destroyed at short times.

Our results show the opposite behaviour in integrable free-fermion chains: Z2 QME is *universal*, occurring for every parameter combination tested. The integrability of the TFIM (and the XY chain family) ensures the existence of quasi-local conserved quantities whose overlap with the initial state persists throughout the dynamics. Different initial polarisation angles θ excite different quasi-particle mode distributions, and the resulting differential relaxation rates produce the Mpemba crossing. The contrast (chaotic $\rightarrow$ Z2 QME absent; integrable $\rightarrow$ Z2 QME universal) mirrors the known contrast for the classical Mpemba effect: the effect is associated with specific non-equilibrium initial conditions that are "remembered" by the dynamics. We conjecture that this dichotomy extends beyond the free-fermion case and constitutes a general feature of integrable versus chaotic symmetry-restoring dynamics.

4.3 Role of dynamical quantum phase transitions

Dynamical quantum phase transitions are among the most prominent non-equilibrium phenomena in isolated quantum systems [8], and it was natural to hypothesise that they might organise the timing of Mpemba crossings. Our statistical analysis provides a nuanced answer to this question.

DQPTs do imprint weakly on the EA dynamics: the distribution of fractional crossing positions ϕ is non-uniform (KS $p \approx 6 \times 10^{-25}$), with crossings tending to occur in the first half of each DQPT period ($\bar{\phi} = 0.377$, peak at $\phi \approx 0.15$). This suggests that the DQPT periodicity modulates the *envelope* of $\Delta S_A(t)$ oscillations — as the Loschmidt echo passes through successive minima, the EA curves of different initial states are differentially affected, preferentially enabling crossings in the rising phase of each Loschmidt oscillation rather than near the cusp. However, DQPTs do not *trigger* crossings: there is no excess concentration near $\phi = 0.5$ (binary test $p = 0.27$), and QME occurs for all 23 post-quench fields tested including eight FM-phase fields where DQPTs are entirely absent.

This finding clarifies which observables are and are not organised by DQPTs. While DQPTs manifest directly in the Loschmidt return rate [7] and can influence two-point order-parameter correlators [17], the EA crossing time t_{M} is primarily a function of initial state geometry, placing it in a distinct universality class from DQPT-sensitive observables.

4.4 Open questions and outlook

Our work opens several directions for future investigation.

Analytical theory. A microscopic theory of t_{M} for the non-Gaussian product states studied here would require extending the quasiparticle picture of EA dynamics [6] to states that break fermion parity. The parity-sector mixing identified in Sec. 2.2 — the initial state being a coherent superposition of the even and odd sectors — may enable an analytical treatment via the Ramond/Neveu-Schwarz sector decomposition. Such a theory could explain both the $\rho = +0.90$ correlation and the $\phi \approx 0.15$ peak of the fractional-position distribution.

Beyond free fermions. The free-fermion structure of the TFIM and XY chain ensures exact solvability but restricts the scope of the results. Extending the analysis to interacting integrable systems (XXZ chain, Hubbard model) and to symmetry classes beyond Z2 (e.g. $\mathbb{Z}_N$ or U(1)) would determine whether the Z2 QME universality observed here is specific to free-fermion dynamics or reflects a broader principle.

Experimental realisations. Transverse-field Ising chains are directly realised in superconducting qubit arrays [9] and trapped-ion quantum simulators, with the ability to prepare product spin states of the form (4) and measure subsystem entanglement via quantum-state tomography. Our exact initial formula $\Delta S_A(0;\theta) = h_{\mathrm{bin}}((1+\cos^{N_A}\theta)/2)$ provides an unambiguous prediction for the pre-quench asymmetry, and the convergence of t_{M} to a finite thermodynamic-limit value at $N \sim 14$ makes the effect accessible at currently achievable system sizes.

Mpemba effect engineering. The strong $\rho = +0.90$ correlation between $|\Delta\Delta\mathcal{S}|$ and t_{M} suggests a practical handle: by tuning the initial polarisation angle, one can engineer the Z2 restoration time over nearly four orders of magnitude without changing any post-quench parameter. This tunability may find applications in quantum state-preparation protocols where the speed of symmetry restoration is a resource.

5 Conclusion

We have presented a comprehensive numerical study of the Z2 quantum Mpemba effect in the one-dimensional transverse-field Ising model and the anisotropic XY chain family. The key contributions of this work are as follows.

We establish, for the first time, that Z2 QME is a *generic and universal* feature of integrable free-fermion dynamics: QME crossings are observed for all 23 post-quench field values tested, spanning the ferromagnetic phase (no DQPTs), the quantum critical point, and the paramagnetic phase (DQPTs present). The 100% occurrence rate across all parameter combinations tested — 69 field-pair combinations in the phase scan and 671 pairs in the statistical analysis — contrasts sharply with the expected absence of Z2 QME in random unitary circuits, highlighting integrability as a key enabler of the effect through long-lived quasiparticle memory.

We identify the *initial entanglement asymmetry imbalance* $|\Delta\Delta\mathcal{S}|$ as the quantitative predictor of the crossing time t_{M}, with a Spearman correlation $\rho = +0.90$ ($p \approx 3 \times 10^{-240}$): a larger initial EA gap produces a later crossing, enabling t_{M} to be tuned over nearly four orders of magnitude purely through initial state preparation. This initial-state origin of t_{M} is made precise by the exact formula $\Delta S_A(0;\theta) = h_{\mathrm{bin}}((1+\cos^{N_A}\theta)/2)$, which provides a machine-precision-verified anchor and saturates the universal Z2 bound $\Delta S_A \leq \ln 2$.

We demonstrate, through rigorous statistical analysis of 671 pairs, that *dynamical quantum phase transitions do not govern QME crossing times*. While the ϕ distribution is non-uniform (KS $p \approx 6 \times 10^{-25}$), indicating that DQPT periodicity modulates the dynamics, the non-uniformity is not centred on the DQPT location (binary test $p = 0.27$), and QME is present even in the DQPT-free ferromagnetic phase. DQPTs and QME thus belong to distinct categories of post-quench observables.

Finally, the crossing times converge smoothly to finite thermodynamic-limit values (confirmed by $1/N$ extrapolation over $N = 8$–18) and are robust with respect to subsystem size ($N_A = 2$–6) and model anisotropy ($\gamma = 0$–1), establishing Z2 QME as a stable bulk phenomenon rather than a finite-size artefact.

These findings open a path toward a predictive theory of symmetry-restoration dynamics in integrable systems, where the initial EA imbalance — directly encodable through initial polarisation angles in cold-atom platforms — writes the timescale of quantum Mpemba crossings.

Supplemental Material

Fig. S1 — Subsystem-size dependence. The robustness of Z2 QME under variation of the subsystem size N_A is demonstrated at $N = 18$, $g_f = 2.0$ for four θ-pairs and $N_A \in \{2,3,4,5,6\}$ (20 combinations total). QME is present in all 20 cases; t_{M} varies with N_A as expected from the scaling of the initial EA gap [Eq. (5) of the main text].

Fig. S2 — QME occurrence map. The occurrence map of QME in the (θ_1, θ_2) plane for $g_f = 2.0$, $N = 12$, $N_A = 4$, showing that QME is present across the full range of initial pairs with $|\Delta\Delta\mathcal{S}| > 0.05$.

Fig. S3 — XY chain anisotropy dependence. Crossing times t_{M} for the anisotropic XY chain at five values $\gamma \in \{0, 0.25, 0.5, 0.75, 1.0\}$ for both $g_f = 2.0$ (PM) and $g_f = 0.42$ (FM/PM depending on $g_c(\gamma) = \sqrt{1-\gamma}/2$). QME is present for all five anisotropies at both fields.

Fig. S4 — Mode occupancy. Quasiparticle mode-occupancy profiles $\langle n_k \rangle(t)$ for selected θ-pairs, illustrating the differential mode excitation that underlies the EA crossing dynamics.

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